INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES TOULOUSE

Advanced Machining
Smart Machining CM 1

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Smart Manufacturing, iIOT, Industry 4.0



https://www.i-scoop.eu/industry-4-0/

- 7 Tech Trends in Smart Manufacturing and Digital Transformation (private company Anasoft) : https://www.youtube.com/watch?v=BK9A8oXJ7tQ
- What Is Industry 4.0 and Smart Manufacturing? https://www.youtube.com/watch?v=EV1Ygw6_rCs

I-Iot Industrial Internet of Things

"The IIoT is the connection of industrial products such as components and/or machines to the internet with wire and wireless connections, increasing value with additional monitoring, analysis and optimization."

V. Alcácer, V. Cruz-Machado / Engineering Science and Technology, an International Journal 22 (2019) 899–919

study the entire process

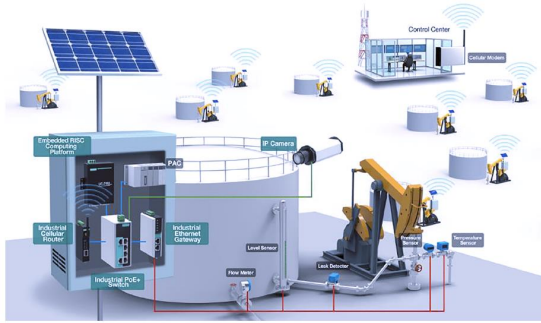


Fig. 4. Typical IIoT network [137].

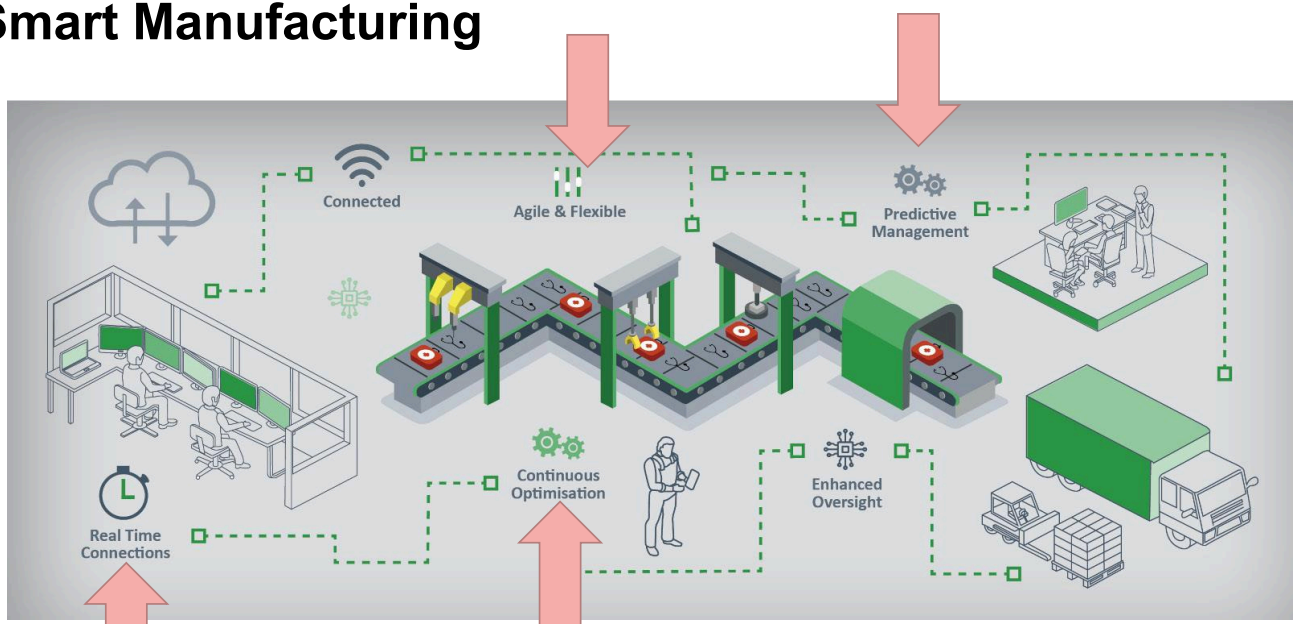
- **Manufacturing operations:** end-to-end visibility, such as planning, production optimization, and supplier management
- **Asset management**: production-asset monitoring, and tracking and monitoring parameters areas, such as quality, performance, potential damage or breakdowns, bottlenecks
- **Field service organizations**: IoT-enabled manufacturing ecosystem
- **Remote monitoring and operation:** optimizes flow, eliminates waste, and avoids unnecessary work in the process inventory to save energy and costs.
- **Condition-based maintenance**: optimize machine availability, minimize interruption, and increase throughput.

BE FLEXIBLE

- **Big data**: Big data can be used to monitor the quality and the makeup of services and enhance the outcome of this aggregated data.
- Cibersecurity : Protection of data -> technology privacy and market competition.

a closed loop.

Smart Manufacturing



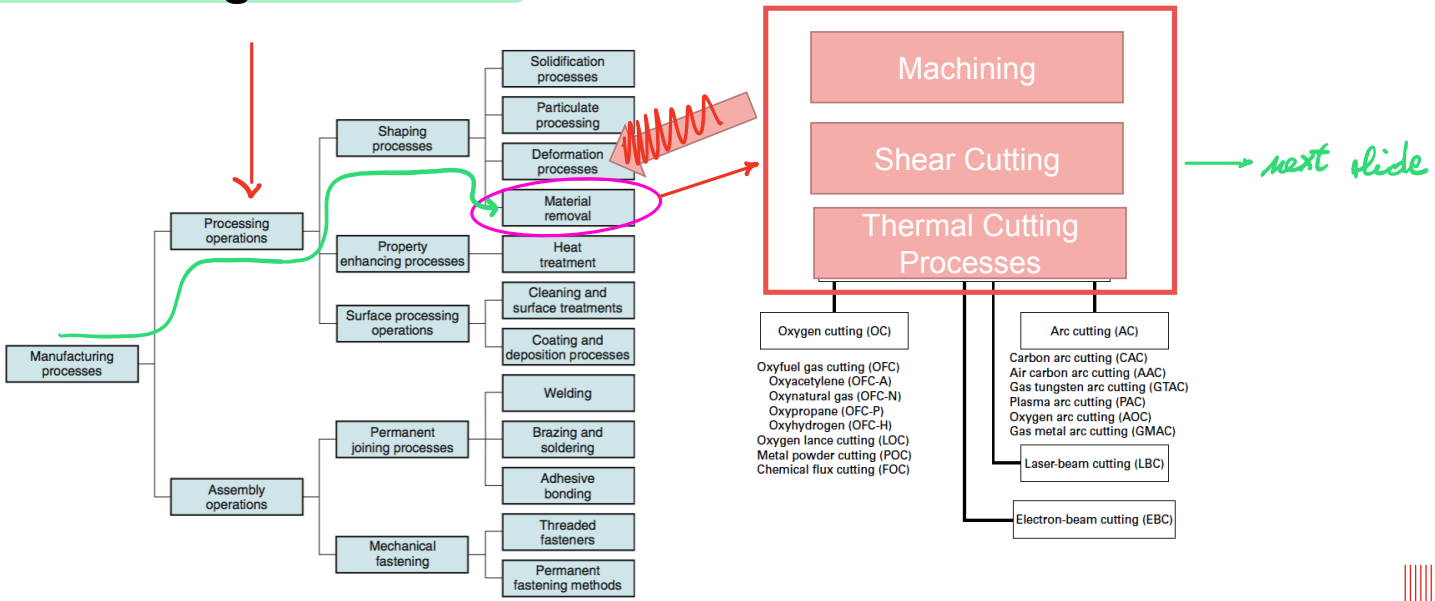
https://slcontrols.com/en/solutions/smart-factory/

Smart Manufacturing

- **Knowledge** of the manufacturing process
- **Monitoring** the proper variables
- **Closed loop** for improving the production or process

Machining Processes

(What we are going to study)



→ next slide

FIGURE 1.3 Classification of manufacturing processes. (Credit: *Fundamentals of Modern Manufacturing*, 4th Edition by Mikell P. Groover, 2010. Reprinted with permission of John Wiley & Sons, Inc.)

Mechanical Machining Processes

- Continious cutting edge processes (Turning, Milling, Drilling, ...)
- Traditional Abrasive Processes (Grinding, Honing, etc)
- Mechanical Energy Processes (Mechanical energy in some form other than the action of a conventional cutting tool. Erosion of the work material by a high velocity stream of abrasives or fluid (or both) is a typical form of mechanical action in these processes.)
  - Ultrasonic Machining
  - Processes Using Water Jets

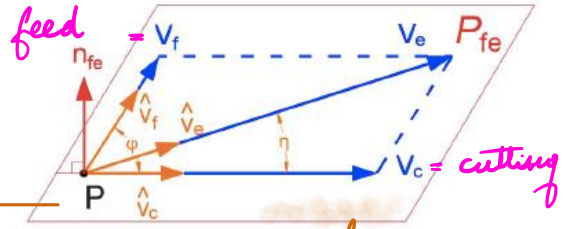
Traditional Machining Processes

Basic Concepts

Machining Motions

removes material

make the effective cut



Primary motion : Main motion that causes relative motion between cutting edge and workpiece and removes material (cutting motion).

create the surface shape

The **feed motion** is added to the primary motion to a repeated or continuous chip removal to create a machined surface.

Effective motion = sum of cutting motion and feed motion

Passive motion : Do not remove material (Positioning, adjust, correction, etc)

Machining Processes – Generating shape with trajectory

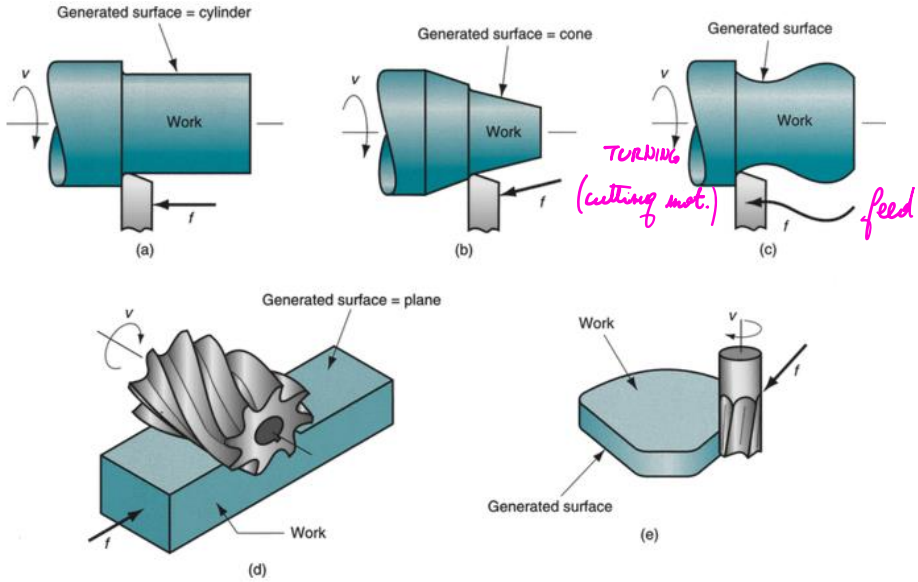


FIGURE 16.2 Generating shape in machining: (a) straight turning, (b) taper turning, (c) contour turning, (d) plain milling, and (e) profile milling. (Credit: *Fundamentals of Modern Manufacturing*, 4th Edition by Mikell P. Groover, 2010. Reprinted with permission of John Wiley & Sons, Inc.)

Machining Processes – Generating shape with tool geometry

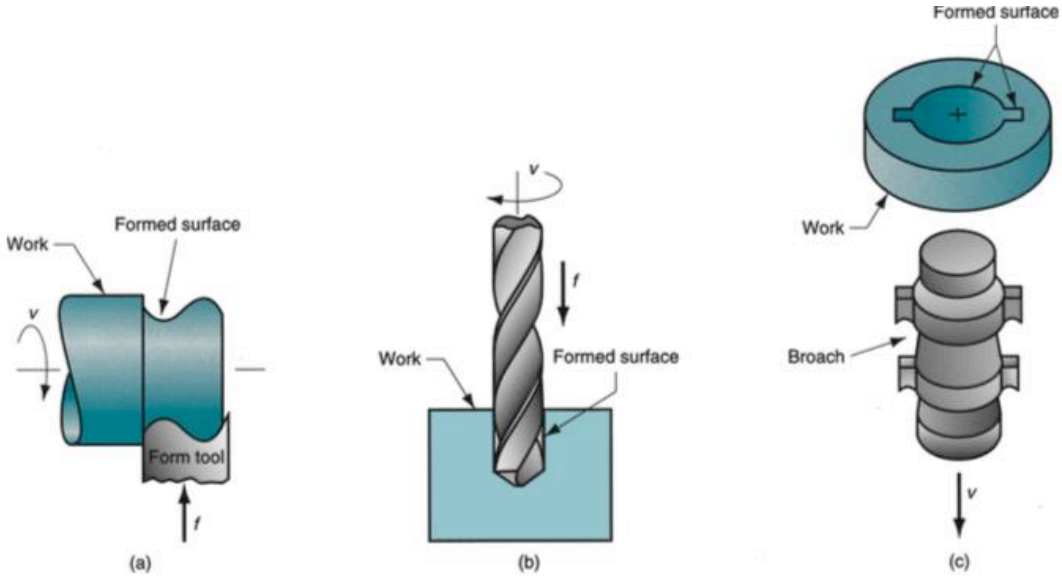
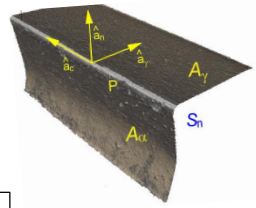
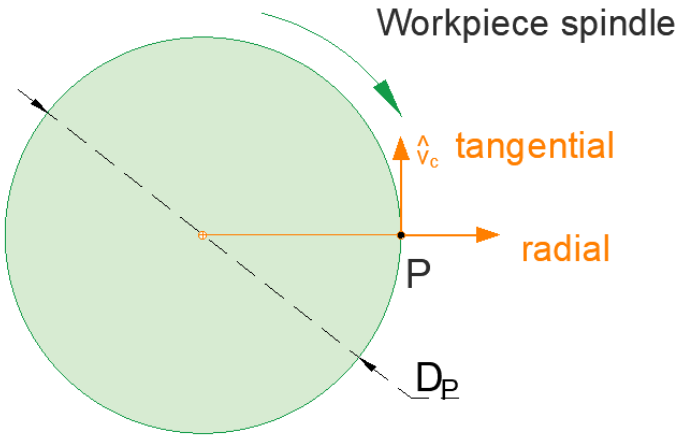


FIGURE 16.3 Forming to create shape in machining: (a) form turning, (b) drilling, and (c) broaching. (Credit: *Fundamentals of Modern Manufacturing*, 4th Edition by Mikell P. Groover, 2010. Reprinted with permission of John Wiley & Sons, Inc.)

Cutting movement Local to the cutting point (Tool or Workpiece turning)



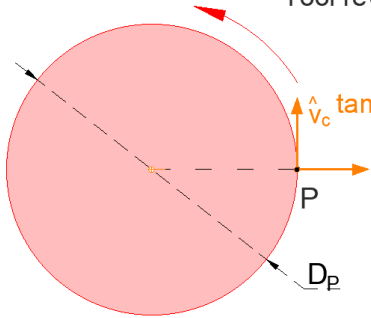


Trajectory of the tool around the workpiece

$$N(\text{rpm}) = \frac{1000\, V_c(\text{m/min})}{2\pi\, R_P(\text{mm})}$$

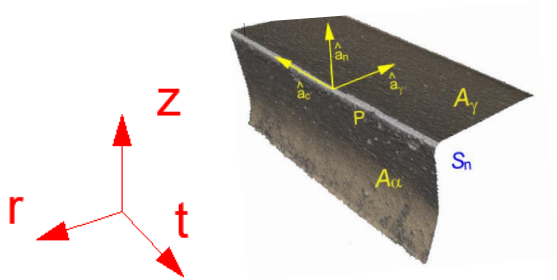
f(DIAMETER)

Tool revolution

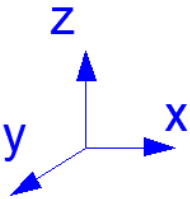


= cutting velocity

Tool envelop







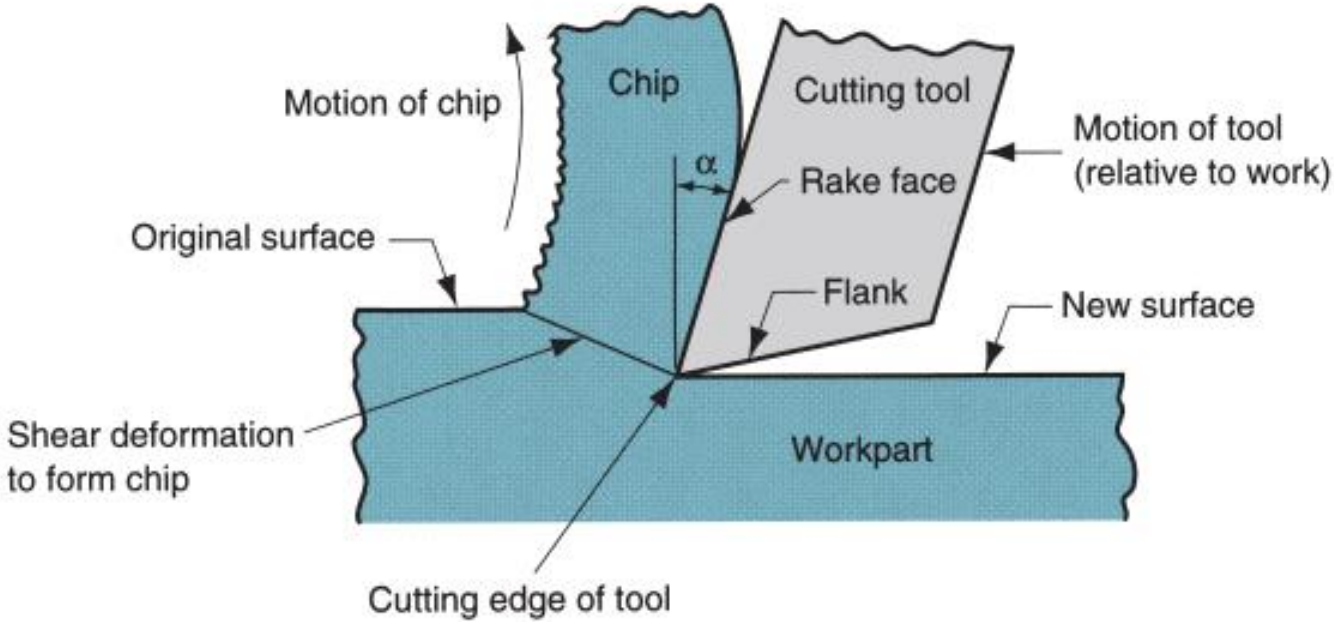
Parameters on referential frame related to the tool envelop surface

Cutting Speed (m/min)	V_c
Feed per cutting edge	f_z (Feed per tooth)
Depth of cut [mm] (and width of cut) a_e	a_p

Parameters on machine tool referential frame

Spindle speed (rpm)	N or S
Feed velocity (mm /min)	V_f

(Feed rate)


Motion of chip
Chip
Cutting tool
Motion of tool
(relative to work)
α
Rake face
Original surface
Flank
New surface
Shear deformation
to form chip
Workpart
Cutting edge of tool

Machining Motions

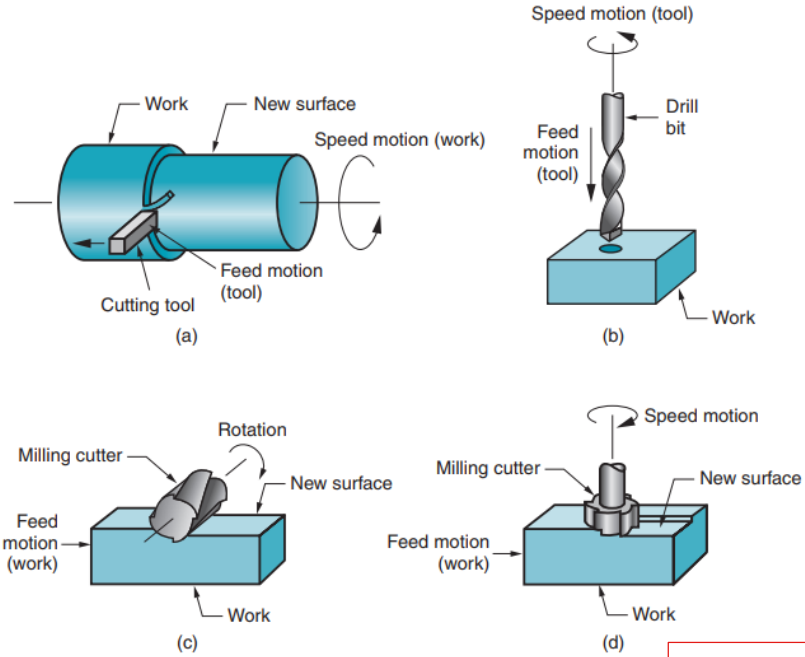


FIGURE 15.2 The three most common types of machining processes: (a) turning, (b) drilling, and two forms of milling: (c) peripheral milling, and (d) face milling. (Credit: *Fundamentals of Modern Manufacturing*, 4th Edition by Mikell P. Groover, 2010. Reprinted with permission of John Wiley & Sons, Inc.)

$$V_f(mm/min) = f_z(mm/th).Z(tooth).N(rpm)$$

$$N(\text{rpm}) = \frac{1000\,V_c(\text{m/min})}{2\pi\,R_P(\text{mm})}$$

Machining on the lathe

Turning



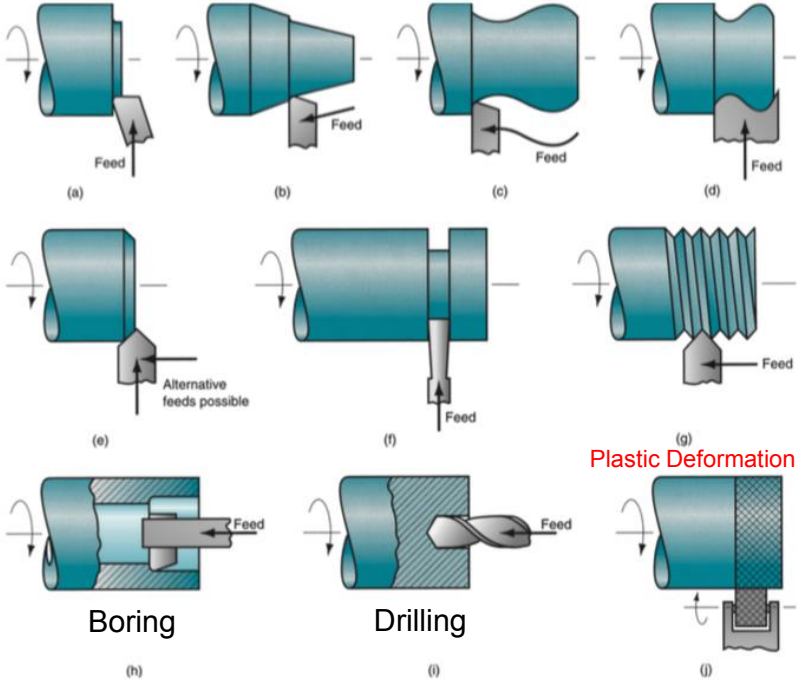
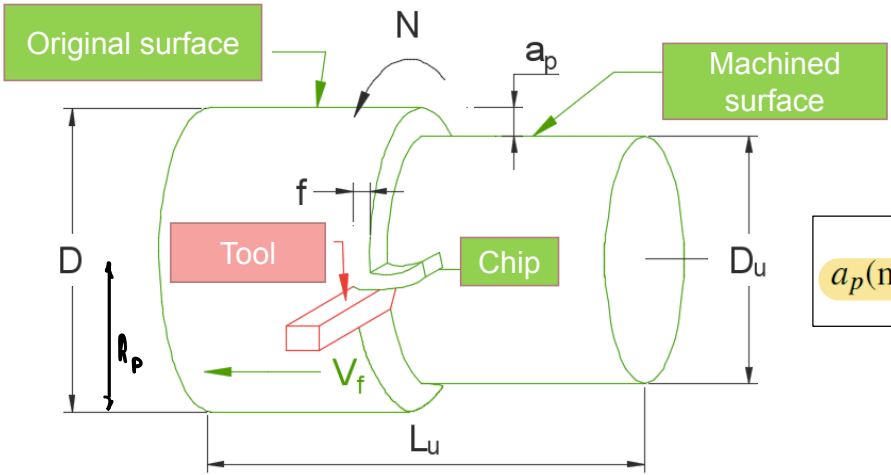


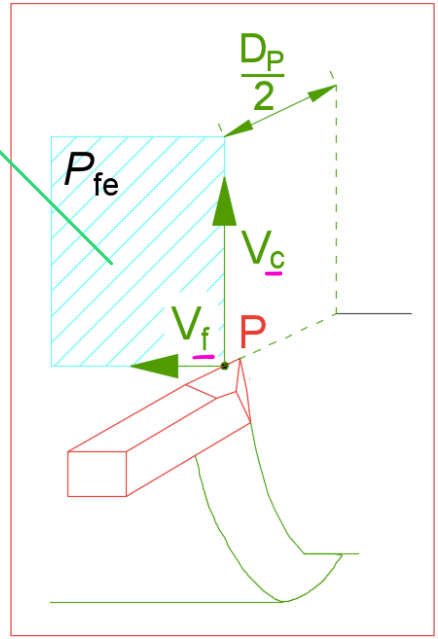
FIGURE 16.6 Machining operations other than turning that are performed on a lathe: (a) facing, (b) taper turning, (c) contour turning, (d) form turning, (e) chamfering, (f) cutoff, (g) threading, (h) boring, (i) drilling, and (j) knurling. (Credit: *Fundamentals of Modern Manufacturing*, 4th Edition by Mikell P. Groover, 2010. Reprinted with permission of John Wiley & Sons, Inc.)

Cylindrical Turning

WORKING PLANE



$$a_p(\text{mm}) = \frac{D - D_u}{2}$$

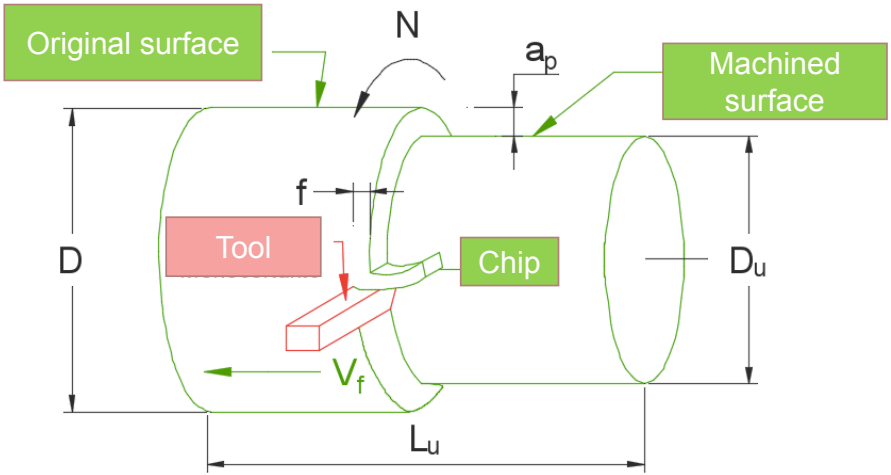


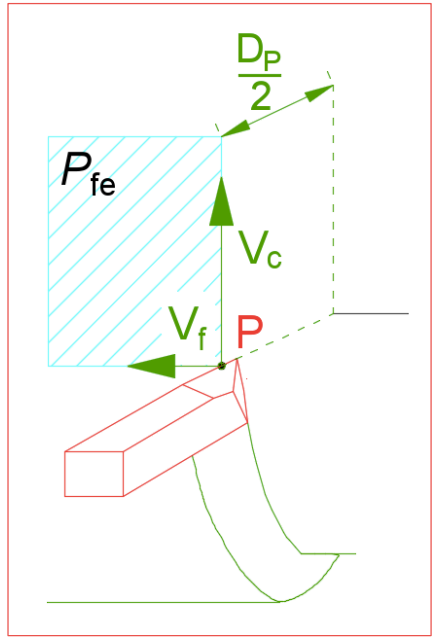
Cutting time (min)

$$N(\text{rpm}) = \frac{1000\, V_c(\text{m/min})}{2\pi\, R_P(\text{mm})}$$

$$t_u(\text{min}) = \frac{L_u(\text{mm})}{V_f(\text{mm/min})}$$

Cylindrical Turning





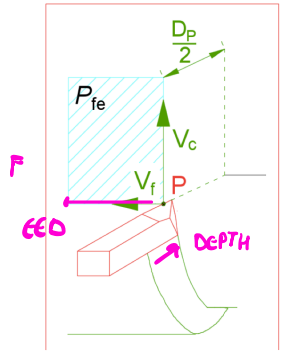
Chip removed volume (mm^3)

$$\mathcal{V}_u(\mathrm{mm}^3) \approx a_p \pi D L_u$$

Chip removal rate (mm^3/min)

$$Q_{rm} = \frac{\pi D a_p L_u}{t_u}$$

Feed and Depth of cut (Area of cut)

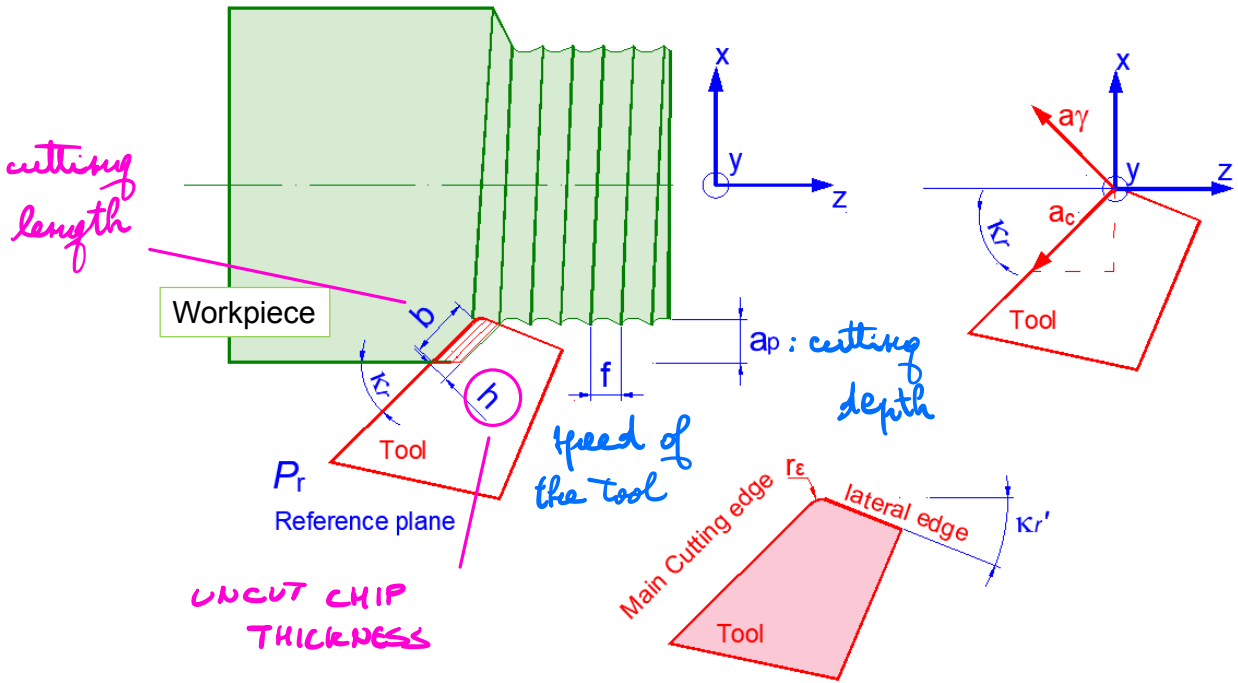


Feed (per cutting edge or per tooth) is the path length **(mm)** that is done to remove a new material part (Feed velocity is the displacement speed of the tool or workpiece (**mm/min)**

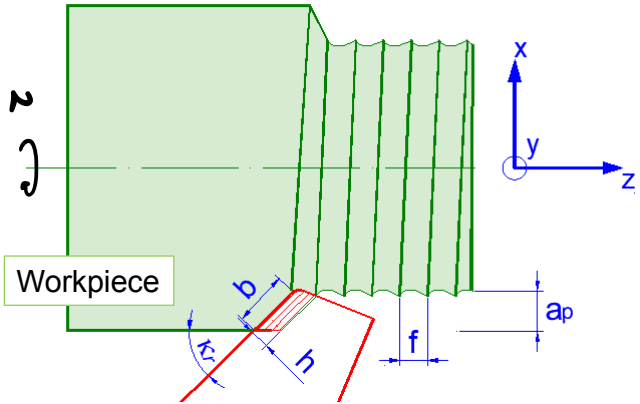
Depth of cut **(mm)** : Vertical penetration of the cutting edge (perpendicular to feed and cutting speed)

Depth of cut **(mm) x Feed (mm)** : Material that the cutting rake face is faced on the cut – It is the quantity of material that resists the tool to be cut

Example of Cylindrical Turning
(Cutting speed = Tangential, Feed = Longitudinal)



Example of Turning (Cutting speed = Tangential)



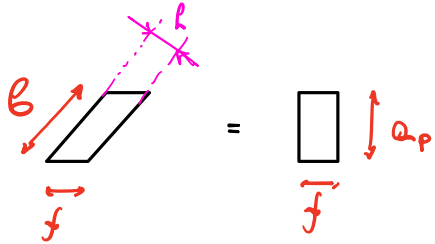
Initial Bar Diameter = 30 mm := D
Ap=1mm
f=0,5 mm/rev ~ relative feed
Area of cut = 0,5 mm²

Cutting Speed = 100 m/min := v_c

What should be the Spindle Speed (rpm) ?

What should be the Feed velocity (mm/min) ?

Area of cut = $b \cdot h = a_p \cdot f$



Spindle speed:

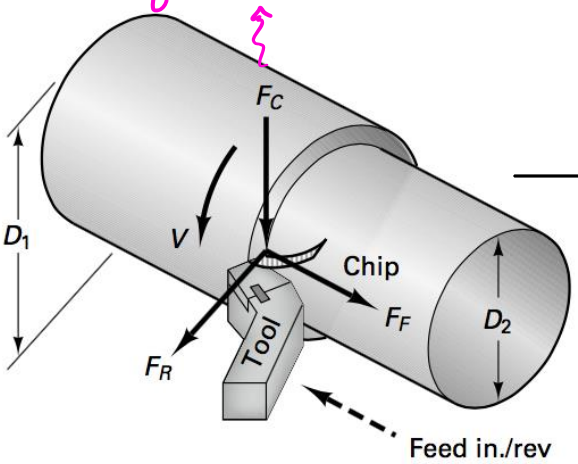
$$N = \frac{v_c}{2\pi \frac{D}{2}} = \frac{100 \text{ m/min}}{\pi \cdot 0.03 \text{ m}} = 1061 \text{ rpm}$$

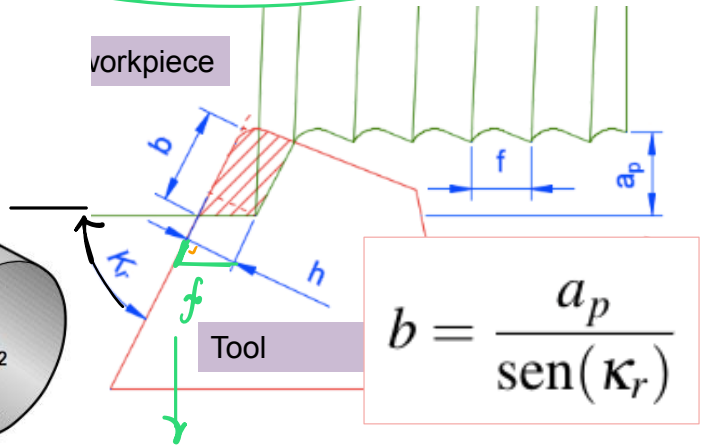
Feed velocity:

$$v_F = f \cdot z_{TOOTH}^{1} \cdot N = 0.5 \frac{\text{mm}}{\text{rev}} \cdot 1061 \frac{\text{rev}}{\text{min}} = 530.5 \frac{\text{mm}}{\text{min}}$$

Cutting area and forces in turning

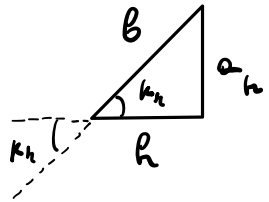
F_S By WORKPIECE on TOOL





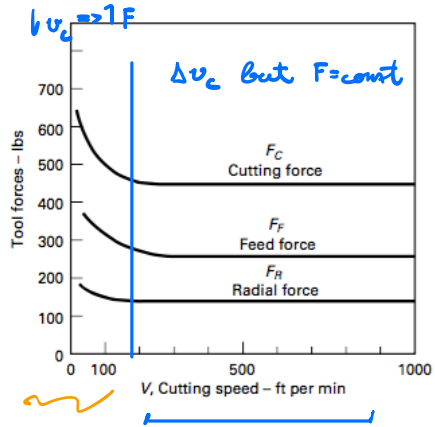
$$b = \frac{a_p}{\text{sen}(\kappa_r)}$$

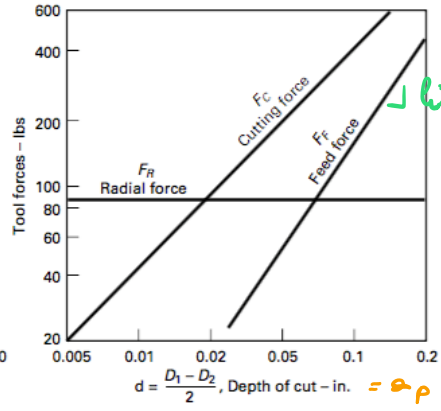
$$h(P) = f\,\text{sen}(\kappa_r)$$

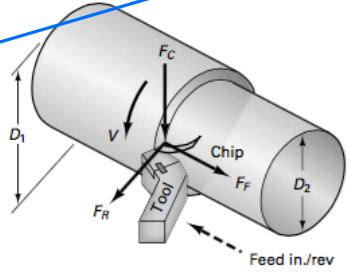


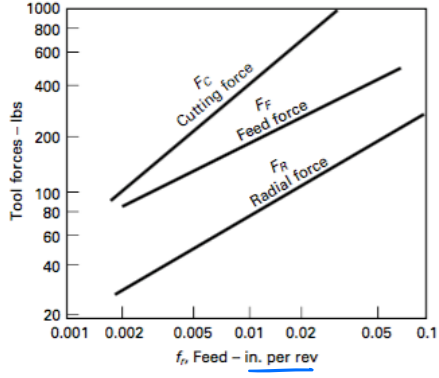
$a_p = b \sin \kappa_r$

Experiments









3 Force

F_C = Cutting force (vertical)

F_R = Radial force (thrust)

F_F = Feed force

Mechanistic Models:

$$F_c = K_c A_c = K_c b h$$

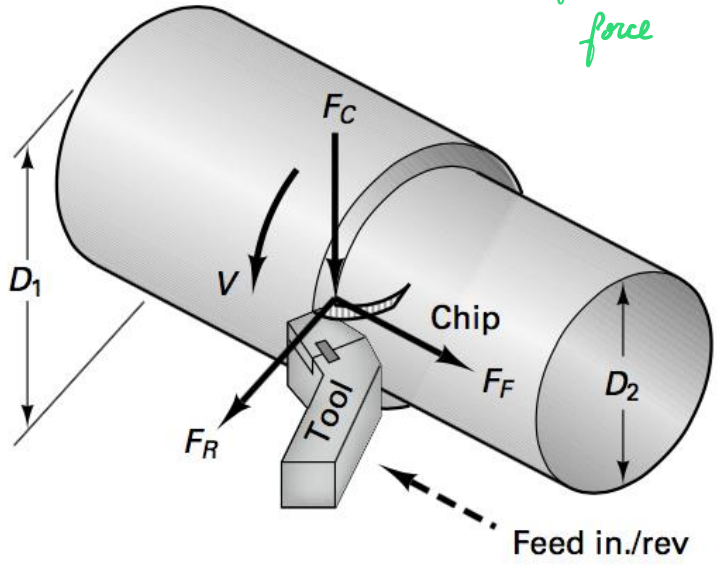
cutting edge dimension

K_c = specific cutting force

$$K_c = function(h, V_c, materials) \left[\frac{N}{m^2}\right]$$

$$F_F = K_F A_c = k_F . F_c$$

$$F_R = K_R A_c = k_R . F_c$$



$$\vec{F}_u = \vec{F}_c + \vec{F}_f + \vec{F}_R$$

Power in Turning

$$P_c = K_c \underbrace{f\, a_p V_c}_{= A_c} \qquad \left[\frac{N}{m^2}\cdot\frac{m}{rev}\cdot m\cdot\frac{m}{s} = \frac{J}{s} = W\right]$$

$$P_f = K_f\, f\, a_p\, V_f\, 10^{-3}$$

↳ UNIT SCALING

$$P_c = \vec{F}_c.\vec{V}_c = F_c V_c = K_c A_c V_c$$

$$P_{req} = \frac{P_u}{\eta_{ef}} \qquad P_c + P_f \approx P_c$$

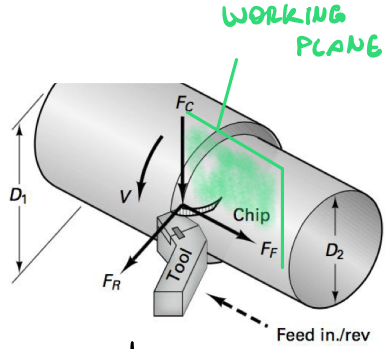
required

$\eta_{ef}\ (0 < \eta_{ef} < 1)$

$$\approx \frac{P_c}{\eta}$$

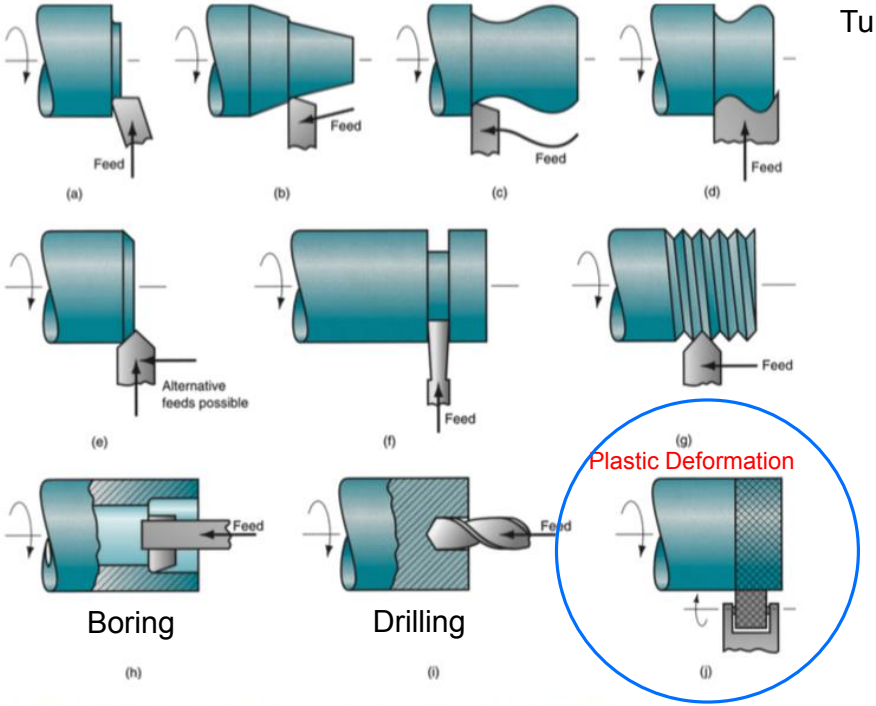
$P_R = 0 \Rightarrow \alpha_R = 0$: $F_R \perp$ quot. ∈ WORK. PLANE

$P_f \ll P_c$



Questa affermazione su Fr NON è vera:
lungo la dir(r) c'è una forza ed una velocità, ma suppongo che sia così ridotta t.c. Pr = 0 ca

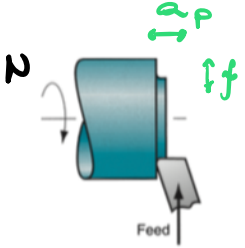
Machining on the lathe



Turning

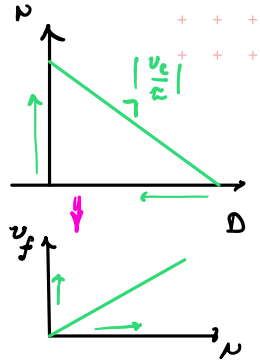
FIGURE 16.6 Machining operations other than turning that are performed on a lathe: (a) facing, (b) taper turning, (c) contour turning, (d) form turning, (e) chamfering, (f) cutoff, (g) threading, (h) boring, (i) drilling, and (j) knurling. (Credit: *Fundamentals of Modern Manufacturing*, 4th Edition by Mikell P. Groover, 2010. Reprinted with permission of John Wiley & Sons, Inc.)

Machining on the lathe



$N = \frac{v_c}{\pi D}$ → 1N ↔ 1D : v_c = cst

N ↑ DECREASING



$v_f = f \cdot N(D) = 530.5$ mm/min

f = 2 mm²

$F_c = K_c \cdot A_c = 700$ MPa $\cdot$ 0.5 = 350 N

$P_c = F_c \cdot v_c = 350\,N \cdot \frac{100}{60}\frac{m}{s} = 583\,W$

Initial Bar Diameter = 30 mm
Ap=1mm
f=0,5 mm/rev
Area of cut = 0,5 mm²

Cutting Speed = 100 m/min
What should be the Spindle Speed (rpm) ?
What should be the Feed velocity (mm/min) ?
If Kc = 700 MPa, what is the cutting force and power ?

Machining Motions

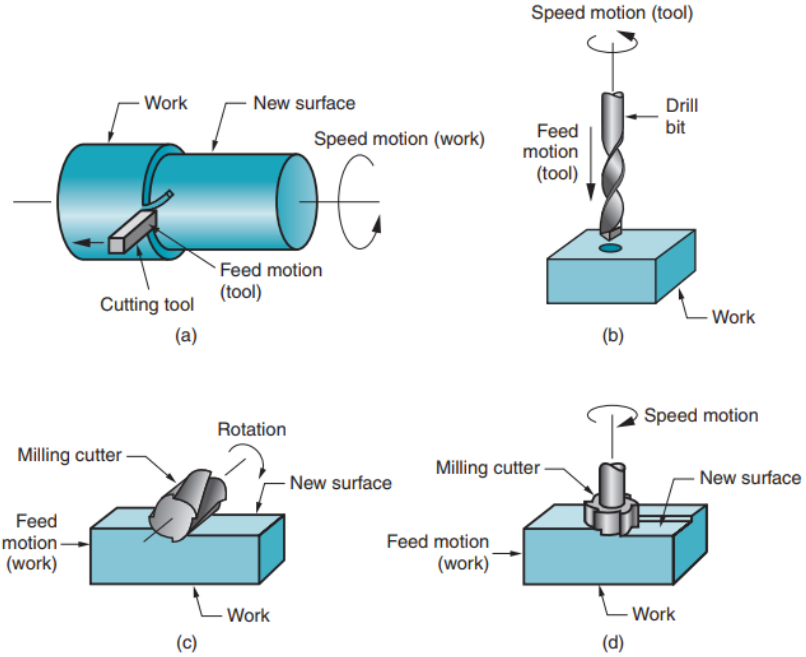
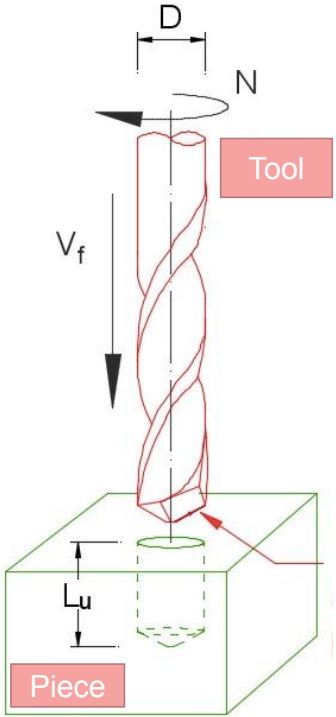


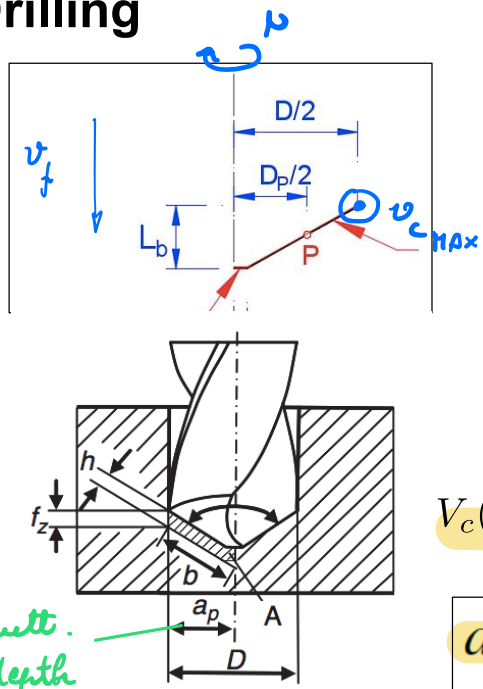
FIGURE 15.2 The three most common types of machining processes: (a) turning, (b) drilling, and two forms of milling: (c) peripheral milling, and (d) face milling. (Credit: *Fundamentals of Modern Manufacturing*, 4th Edition by Mikell P. Groover, 2010. Reprinted with permission of John Wiley & Sons, Inc.)

$$N(\text{rpm}) = \frac{1000\, V_c(\text{m/min})}{2\pi\, R_P(\text{mm})}$$

$$V_f(mm/min) = f_z(mm/th).Z(tooth).N(rpm)$$

Cutting Area in Axial Drilling





$$= \omega r = \omega \cdot \frac{D}{2}$$

$$= 2\pi N \cdot \frac{D}{2}$$

$$V_c max(m/min) = \frac{\pi . N . D}{1000}$$

$$V_c(m/min)(D_P) = \frac{\pi . N . D_P}{1000}$$

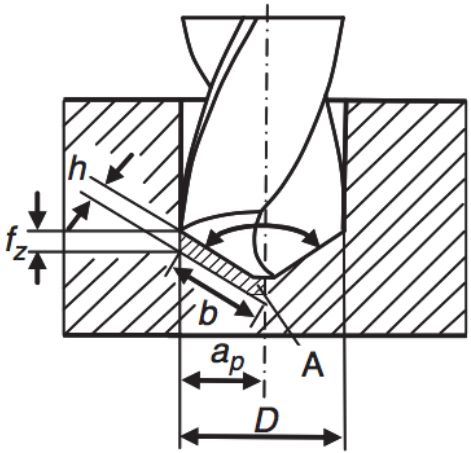
scale factor

$$a_p = \frac{D}{2}$$

$$\vec{V}_f(\text{mm/min}) = f\,N = f_z\,z\,N = 2\,f_z\,N$$

tooth

Machining Forces in Drilling



Values per tooth:

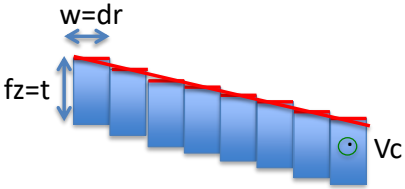
Fc1 = Kc. ap . fz → force single tooth

Pc1 = Fc. (Vc/2) → not the total: $\underline{F}_{tot} = \underline{0}$

mean of v_c : it varies along the edge

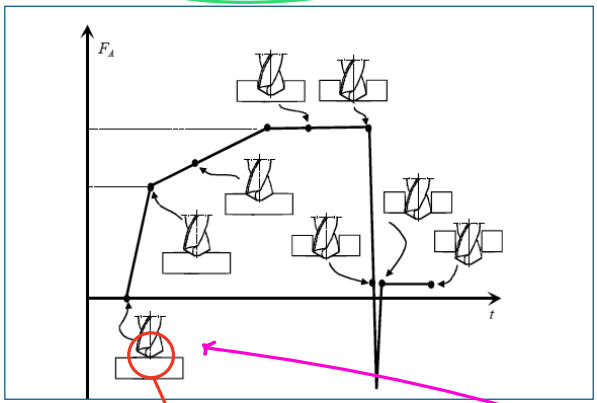
We can calculate the Cutting force per tooth.
But when we make the sum, the resultant is zero.

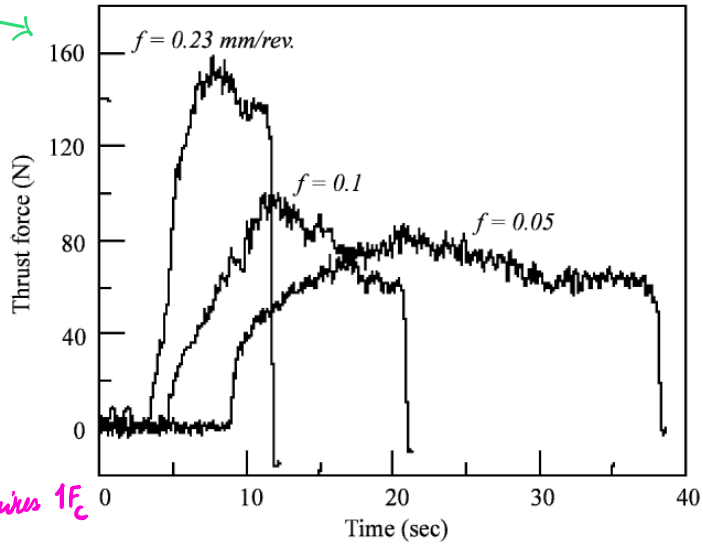
Although, the power of both flutes summed it not zero !
and also the Torque ≠ 0 !



Machining Forces in Drilling – Axial Force in Z direction

Vertical (Axial) Force





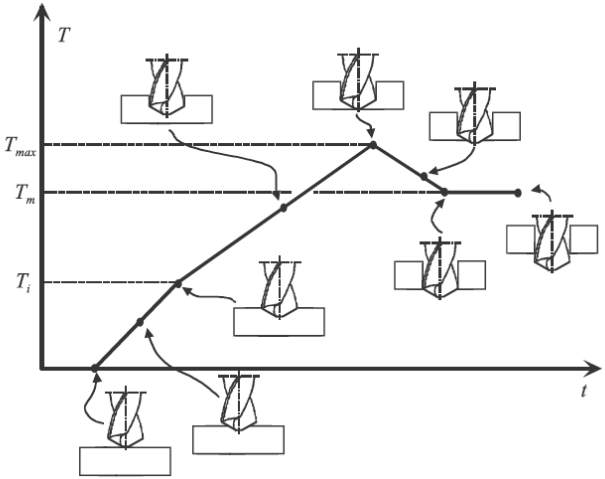
"Machining - Fundamentals and Recent Advances – Davim, J., 2008 – Ed. Springer

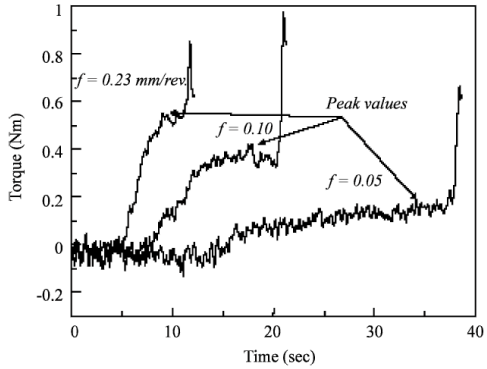
it requires $1F_c$

drilling heads are manufactured to make a PLASTIC DEF, instead of a CUT ⇝ because $v_c(r=0) = 0 \Rightarrow F_c = 0$

"Machining - Fundamentals and Recent Advances – Davim, J., 2008 – Ed. Springer

Torque in Drilling





"Machining - Fundamentals and Recent Advances – Davim, J., 2008 – Ed. Springer

Exercise :

Choose your tool, and use the same parameters that we used before :
f =0,5 mm/rev
Kc=700 MPa
Vc=100m/min
And calculate Cutting Force Fc and Cutting Power (after the point is already inside the material)

Tool entered (plateau of F_A) :

$$a_p = \frac{D}{2} \rightarrow F_{c_1} = K_c \cdot a_p \cdot f$$

$$\downarrow$$

$$P_{c_2} = F_{c_1} \cdot \frac{v_c}{2}$$

Exercise :

What is the variation of the cutting speed along the edge ?
What is your prediction of force from 0 to the depth of the hole = 3 times the diameter of the tool ?
Make a curve with the prediction of the power along the time in seconds.